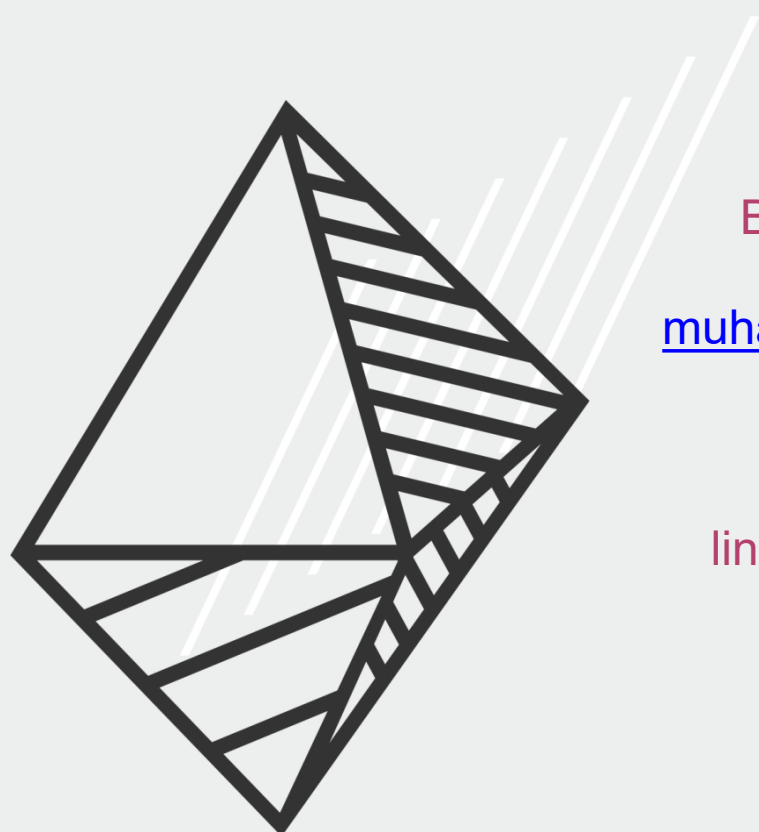


EAssess

A Secure and Scalable Platform for
Online Examinations

Software Requirement Specification (SRS)



By : Muhammed Salah Hussain

muhammed.salah.hussain@gmail.com

github.com/M-S-H-Git

linkedin.com/in/muhammed-salah-hussain-231797388

July 2025



1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for the Online Examination System for the Sindh Board of Education (SBoE). This system will facilitate the conduct of entry exams for various institutions through an online platform.

1.2 Document Conventions

This document follows the IEEE 830 standard for Software Requirements Specifications.

1.3 Intended Audience and Reading Suggestions

The intended audience for this document includes:

- Project Managers
- Software Developers
- Quality Assurance Team

Stakeholders from Total-Solutions (TS) and Sindh Board of Education (SBoE)

1.4 Project Scope

The Online Examination System will include:

- A back-end module for entering evaluation questions.
- A front-end module for rendering exams to students.
- An evaluation calculator for marks calculation.
- Integration with SBoE's API for data exchange.

1.5 References

IEEE 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

2. Overall Description

2.1 Product Perspective

The Online Examination System is a standalone web-based application that will integrate with SBoE's existing systems through an API. It will provide a centralized platform for managing and conducting online exams.

2.2 Product Features

- User Authentication and Authorization
- Question Bank Management
- Exam Creation and Management
- Student Exam Interface
- Evaluation and Grading
- Reporting and Analytics

2.3 User Classes and Characteristics

- Administrators: Manage the system, create exams, and enter questions.
- Teachers/Evaluators: Review and approve questions, evaluate exams.
- Students: Take exams and view results.
- IT Staff: Maintain the system and ensure API integration.

2.4 Operating Environment

- Web-based application accessible via standard web browsers.
- Compatible with desktop and mobile devices.

2.5 Design and Implementation Constraints

- Compliance with data protection regulations.
- Secure communication with SBoE's API.
- Scalability to handle many concurrent users.

2.6 User Documentation

- User manuals for administrators, teachers, and students.
- Online help and tutorials.

2.7 Assumptions and Dependencies

- Reliable internet connection for all users.
- Availability of SBoE's API for integration.

3. System Features

3.1 User Authentication and Authorization

- Description: Secure login and role-based access control.
- Priority: High
- Functional Requirements:
 - FR1.1: The system shall allow users to register and log in.
 - FR1.2: The system shall implement role-based access control.
 - FR1.3: The system shall allow administrators to create and manage user roles with different privileges.
 - FR1.4: The system shall allow administrators to associate roles with privileges (e.g., Add, Modify, Delete).

3.2 Question Bank Management

- Description: Manage a repository of exam questions.
- Priority: High
- Functional Requirements:
 - FR2.1: The system shall allow administrators to add, edit, and delete questions.
 - FR2.2: The system shall support importing questions from Excel files.
 - FR2.3: The system shall allow evaluators to categorize questions by subject and difficulty level.
 - FR2.4: The system shall support various answer types, including multiple-choice, fill-in-the-blanks, and text answers.
 - FR2.5: The system shall restrict evaluators from leaving the question text empty.

3.3 Exam Creation and Management

- Description: Create and manage online exams.
- Priority: High
- Functional Requirements:
 - FR3.1: The system shall allow administrators to create exams by selecting questions from the question bank.
 - FR3.2: The system shall allow scheduling of exams.
 - FR3.3: The system shall allow teachers to associate administrative information (e.g., date, venue) with each exam.
 - FR3.4: The system shall allow the implementation of examination rules (e.g., time limits, question randomization).
 - FR3.5: The system shall track the status of exams (e.g., New, Ready, Expired).
 - FR3.6: The system shall prevent deletion of questions that are already associated with any exam.

3.4 Student Exam Interface

- Description: Interface for students to take exams.

- Priority: High
- Functional Requirements:
 - FR4.1: The system shall provide a user-friendly interface for students to take exams.
 - FR4.2: The system shall display questions one at a time or all at once based on exam settings.
 - FR4.3: The system shall allow students to register and make payments online via credit cards or online bank transfers.
 - FR4.4: The system shall enable exams for students on the scheduled date and time.
 - FR4.5: The system shall enforce system requirements checks before starting the exam.
 - FR4.6: The system shall display animations for correct and incorrect answers during the exam.

3.5 Evaluation and Grading

- Description: Automated evaluation and grading of exams.
- Priority: High
- Functional Requirements:
 - FR5.1: The system shall automatically evaluate multiple-choice questions.
 - FR5.2: The system shall allow evaluators to grade subjective questions.
 - FR5.3: The system shall post student exam results to the SBoE API, including student serial number, exam ID, date of exam, date of submission, and student score.
 - FR5.4: The system shall implement a timeout mechanism for API communication and handle acknowledgements.
 - FR5.5: The system shall display a thank-you message to students after completing the exam.

3.6 Reporting and Analytics

- Description: Generate reports and analytics on exam performance.
- Priority: Medium
- Functional Requirements:
 - FR6.1: The system shall generate reports on student performance.
 - FR6.2: The system shall provide analytics on exam statistics.
 - FR6.3: The system shall allow exporting reports in various formats (PDF, Excel).

3.7 Wireframes

Here are the basic wireframes for the Question/Answer module's workflows, showing the basic structure of pages and how they will be interlinked:

- Wireframe 1: Question Bank Management

Question Bank Management

Question Bank Management

[Add Question] [Import from Excel] [Search]

Question List:

- Question 1
- Question 2
- Question 3
- ...

[Edit] [Delete]

- Description:

Title: Question Bank Management

Buttons: [Add Question], [Import from Excel], [Search]

Question List: Displays a list of questions (e.g., Question 1, Question 2, Question 3, etc.)

Actions: [Edit], [Delete] buttons for each question

- Wireframe 2: Exam Creation

Exam Creation

Exam Creation

Exam Title: [_____]
Exam Date: [_____]
Duration: [_____] minutes

Select Questions:

[] Question 1

[] Question 2

[] Question 3

...

[Save Exam] [Cancel]

- Description:

Title: Exam Creation

Fields:

Exam Title: [_____]

Exam Date: [_____]

Duration: [_____] minutes

Question Selection:

Question 1

Question 2

Question 3

...

Buttons: [Save Exam], [Cancel]

- Wireframe 3: Add Question

Add Question

Add Question

Question Type: [Dropdown]

If MCQ selected:

Question Text: [_____]

Answer Choices:

[] Choice A [] Choice B [] Choice C [] Choice D

If Fill in the Blanks selected:

Question Text: [_____] [Blank]

Correct Answer: [_____]

If Text Answer selected:

Question Text: [_____]

Answer: [_____]

- Description:

Title: Add Question

Dropdown: Question Type: [Dropdown]

MCQ Type:

Question Text: [_____]

Answer Choices: [] Choice A [] Choice B [] Choice C [] Choice D

Fill in the Blanks Type:

Question Text: [_____] [Blank]

Correct Answer: [_____]

Text Answer Type:

Question Text: [_____]

Answer: [_____]

These wireframes provide a basic structure of the pages and how they will be interlinked, helping stakeholders understand the flow of the Question/Answer module.

4. External Interface Requirements

4.1 User Interface

- UI1.1: The system shall provide a web-based interface accessible via standard web browsers.
- UI1.2: The interface shall be responsive and compatible with different screen sizes.

4.2 Hardware Interface

- HW1.1: The system shall support integration with barcode scanners for student ID verification.

4.3 Software Interface

- SW1.1: The system shall integrate with SBoE's API for data exchange.
- SW1.2: The system shall support importing questions from Excel files.

4.4 Communication Interface

- CI1.1: The system shall support secure communication protocols (e.g., HTTPS) for data transmission.

5. Non-Functional Requirements

5.1 Performance Requirements

- NFR1.1: The system shall support up to 1000 concurrent users without performance degradation.
- NFR1.2: The system shall provide a response time of less than 2 seconds for all user interactions.

5.2 Security Requirements

- NFR2.1: The system shall comply with data protection regulations.
- NFR2.2: The system shall implement role-based access control (RBAC) to restrict access to sensitive data.
- NFR2.3: The system shall encrypt data in transit and at rest.

5.3 Software Quality Attributes

- NFR3.1: The system shall have an uptime of 99.9%.
- NFR3.2: The system shall provide user guides and tutorials for new users.

6. Development Strategy

6.1 Feature Priority Assessment

Based on the requirements gathered and the expectations/limitations from Total-Solutions (TS), the features can be prioritized as follows:

- User Authentication and Authorization (High Priority)
- Question Bank Management (High Priority)
- Exam Creation and Management (High Priority)
- Student Exam Interface (High Priority)
- Evaluation and Grading (High Priority)
- Integration with SBoE's API (High Priority)
- Reporting and Analytics (Medium Priority)

6.2 Iteration Plan

Given the resource situation and the need to demonstrate a working first iteration within one month, the features will be implemented in the following iterations:

- Iteration 1 (First Month)
 - User Authentication and Authorization: Implement secure login and role-based access control.
 - Question Bank Management: Develop the functionality to add, edit, delete, and import questions.
 - Exam Creation and Management: Enable the creation of exams by selecting questions from the question bank and scheduling exams.
 - Integration with SBoE's API: Ensure the system can communicate with SBoE's API for data exchange.
- Iteration 2 (Second Month)
 - Student Exam Interface: Develop the interface for students to register, make payments, view exams, and take exams.
 - Evaluation and Grading: Implement automated evaluation of multiple-choice questions and manual grading of subjective questions.
 - System Requirements Check: Ensure the system checks for required software/configurations on student systems before starting the exam.
- Iteration 3 (Third Month)
 - Reporting and Analytics: Generate reports on student performance and provide analytics on exam statistics.
 - Additional Features: Implement any additional features or enhancements based on feedback from the first two iterations.

6.3 Additional Features for First Iteration

To ensure a successful first iteration and meet the expectations of TS, the following additional features may be considered:

- Basic Exam Interface: Develop a simple mock exam interface to demonstrate the flow of questions and basic evaluation.
- API Communication Demonstration: Create a test module to demonstrate successful communication with SBoE's API.
- Feedback Mechanism: Add a simple feedback mechanism for users to provide feedback on the system.
- System Requirements Check: Implement basic system requirements check to ensure that the student's system meets the necessary software and configuration requirements.

7. Appendix

7.1 Appendix A: Analysis Models

- Use Case Diagrams:

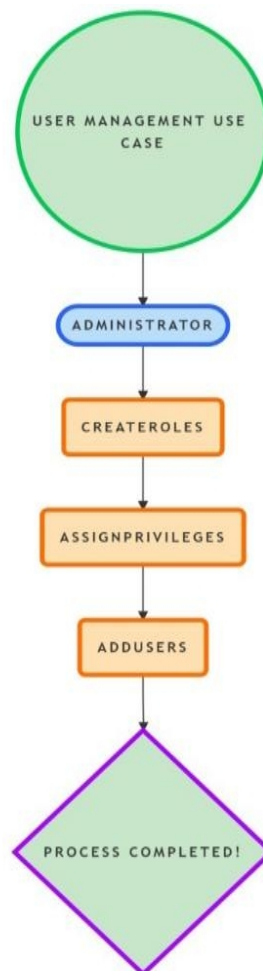


Figure 1 : User Management

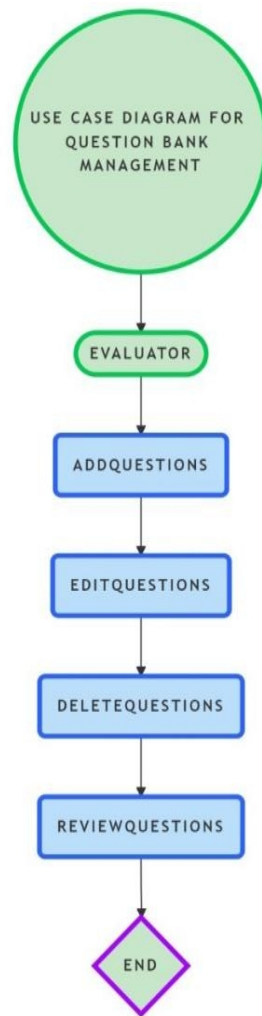


Figure 2 : Question Bank Management

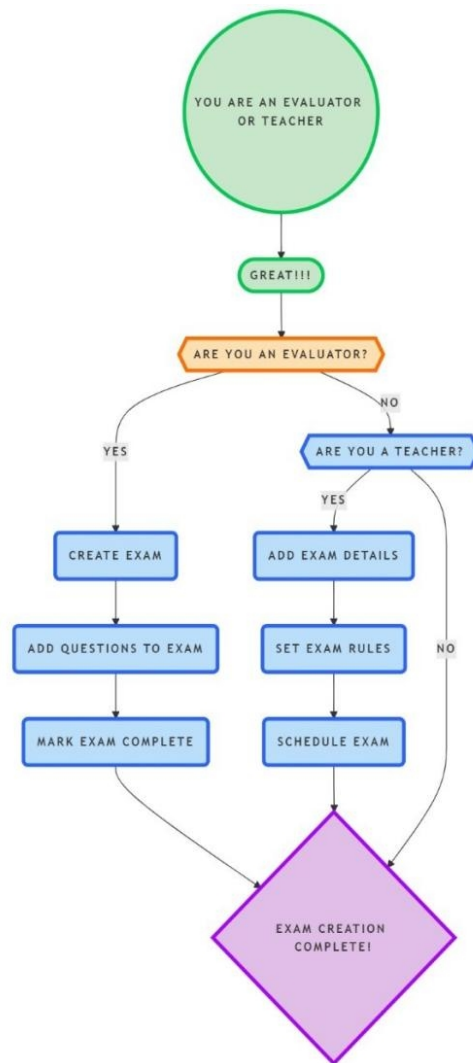


Figure 3 : Evaluation

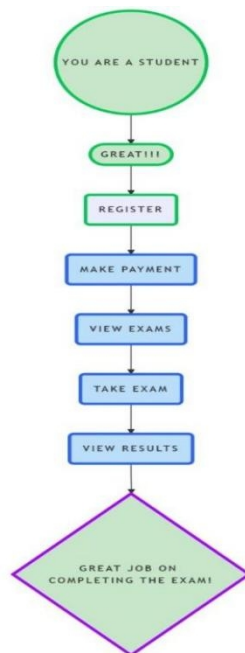


Figure 4 : Student

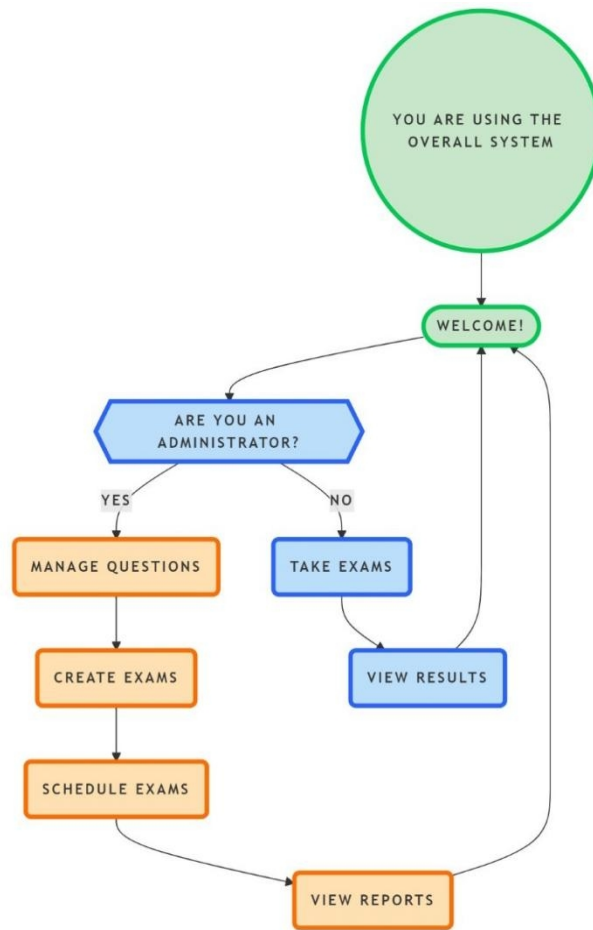


Figure 5 : Overall System

- Activity Diagrams: The following activity diagrams provide a visual representation of the workflows for key processes in the Online Examination System, helping to clarify the steps involved in exam creation, student exam interface, and question management.

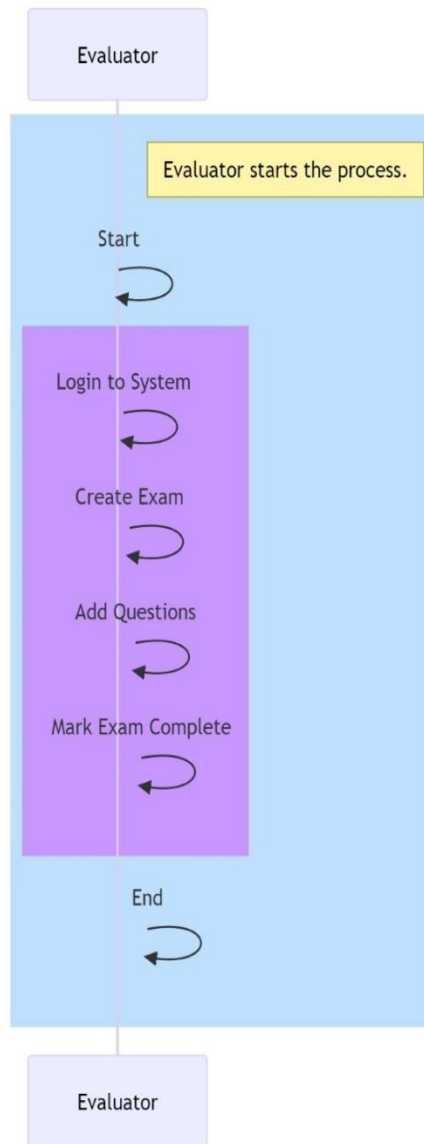


Figure 6 : Activity Diagram for Exam Creation

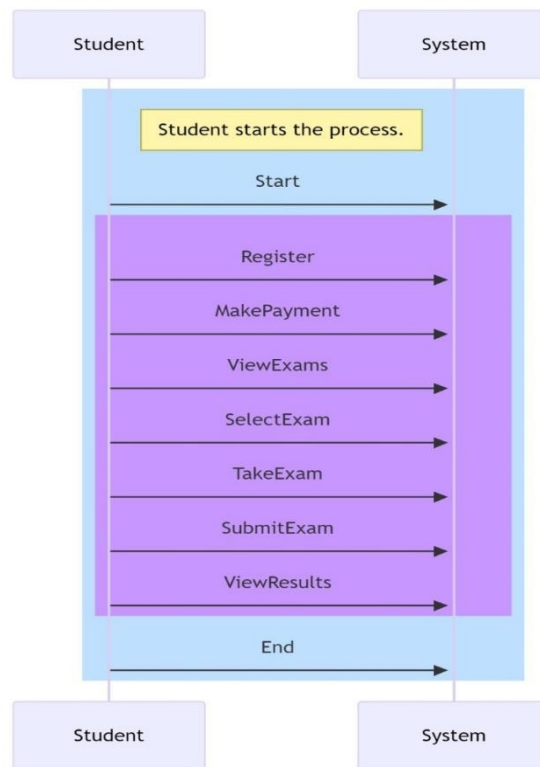


Figure 7 : Activity Diagram for Student Exam Interface

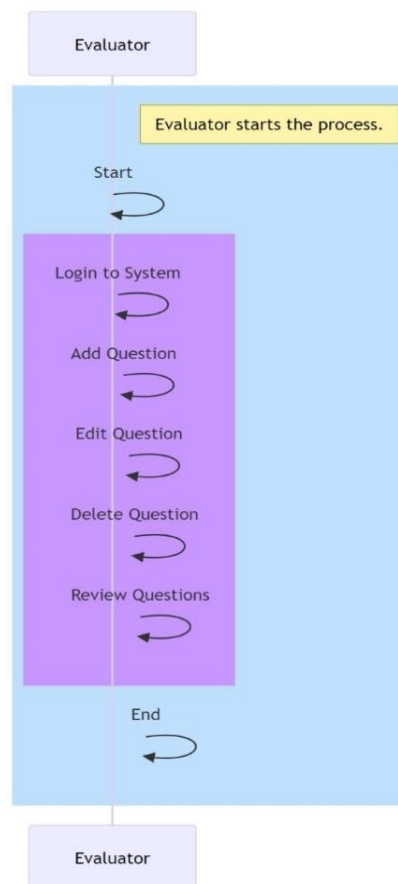


Figure 8 : Activity Diagram for Question Management

7.2 Appendix B: Issue List

This section lists the issues identified during the requirement gathering and analysis phase for the Online Examination System. These issues need to be addressed to ensure the successful implementation and operation of the system.

- Issue 1: Integration with SBoE's API
 - Description: Ensuring successful communication with SBoE's API is critical. Some competitors were unable to communicate with it, leading to their bids being rejected.
 - Impact: High
 - Resolution: Conduct research and develop a test module to demonstrate successful API communication. Implement a timeout mechanism and handle acknowledgements from the API.
- Issue 2: Scalability to Handle Concurrent Users
 - Description: The system must support up to 1000 concurrent users without performance degradation.
 - Impact: High
 - Resolution: Optimize the system architecture and perform load testing to ensure scalability.
- Issue 3: Data Security and Privacy
 - Description: The system must comply with data protection regulations and ensure the security and privacy of user data.
 - Impact: High
 - Resolution: Implement role-based access control (RBAC), encrypt data in transit and at rest, and conduct regular security audits.
- Issue 4: System Availability
 - Description: The system needs to be reliable and always available, especially during examination periods (Jan-Mar and Sept-Dec).
 - Impact: High
 - Resolution: Ensure an uptime of 99.9% and implement backup and recovery procedures.
- Issue 5: User Interface Usability
 - Description: The user interface must be user-friendly and intuitive for all user roles (administrators, evaluators, teachers, students).
 - Impact: Medium
 - Resolution: Conduct usability testing and gather feedback from users to improve the interface design.
- Issue 6: Exam Rules Implementation
 - Description: The system must support the implementation of examination rules (e.g., skipping questions, automatic question skip, moving back).
 - Impact: Medium

- Resolution: Develop a rules engine to enforce the specified rules during exams.
- Issue 7: System Requirements Check for Students
 - Description: The system must check for required software/configurations on student systems before starting the exam.
 - Impact: Medium
 - Resolution: Implement system requirements check mechanism and provide clear instructions to students for installing necessary software.
- Issue 8: Question Repository Management
 - Description: Questions in the repository that are already associated with any exam (active or expired) cannot be deleted.
 - Impact: Medium
 - Resolution: Implement restrictions to prevent the deletion of associated questions and ensure data integrity.
- Issue 9: Reporting and Analytics
 - Description: The system must generate reports on student performance and provide analytics on exam statistics.
 - Impact: Medium
 - Resolution: Develop basic reporting features in the initial iterations and enhance them based on user feedback.

By addressing these issues, we can ensure the successful development and deployment of the Online Examination System, meeting the requirements and expectations of Total-Solutions (TS) and the Sindh Board of Education (SBoE).

Appendix C: Glossary

This glossary provides definitions for terms and acronyms used in the Software Requirements Specification (SRS) for the Online Examination System.

- API: Application Programming Interface. A set of functions and procedures allowing the creation of applications that access the features or data of an operating system, application, or other service.
- RBAC: Role-Based Access Control. A method of regulating access to computer or network resources based on the roles of individual users within an enterprise.
- SBoE: Sindh Board of Education. The educational board for the Sindh region responsible for overseeing educational standards and conducting examinations.
- TS: Total-Solutions. The company contracted to develop the Online Examination System for the Sindh Board of Education.
- HOMS: Hospital Operation Management System. A system designed to streamline hospital operations, including patient management, appointment scheduling, billing, and reporting.
- EMR: Electronic Medical Records. Digital versions of patients' paper charts, containing medical and treatment histories.

- EHR: Electronic Health Records. Digital records of patients' overall health, including medical history, diagnoses, medications, treatment plans, immunization dates, allergies, radiology images, and laboratory test results.
- HIPAA: Health Insurance Portability and Accountability Act. A US law designed to provide privacy standards to protect patients' medical records and other health information.
- MCQ: Multiple-Choice Question. A type of question where the respondent selects one or more answers from a list of choices.
- CMS: Content Management System. A software application used to create, manage, and modify digital content.
- LMS: Learning Management System. A software application for the administration, documentation, tracking, reporting, automation, and delivery of educational courses, training programs, or learning and development programs.
- DB: Database. An organized collection of data, generally stored and accessed electronically from a computer system.
- GUI: Graphical User Interface. A type of user interface that allows users to interact with electronic devices through graphical icons and visual indicators.
- HTTPS: Hypertext Transfer Protocol Secure. An extension of HTTP used for secure communication over a computer network.
- TBD: To Be Determined. Used when a piece of information is not yet available or decided.